

DAT 5564: Database Design and SQL

Fall 2025, Mini A

Lab #5: SQL Part 4

INSTRUCTIONS

1. This is a group assignment to be completed during the lab.
2. Please include only your Student IDs on the submission.
 - a. This submission will be submitted online on Canvas.
 - b. Each member of your group needs to submit, even though you are working in a group.
 - c. This submission is automatically graded.
 - d. The submission is due on Canvas.

SESSION OUTLINE

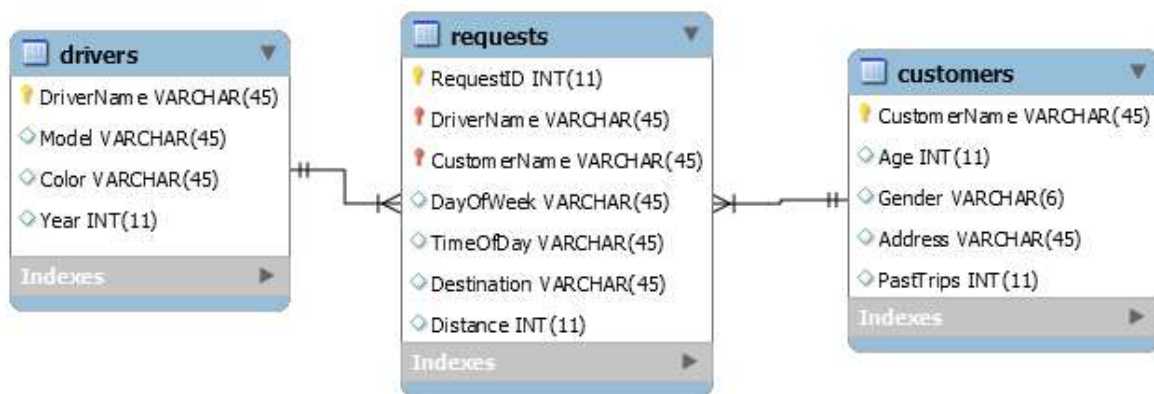
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|---------------------------------|------------|
| • Discuss Questions | 10 minutes |
| • Assignment Questions #1 – #13 | 45 minutes |
| • Solutions to some questions | 25 minutes |

Assignment

Use the Lyft database to answer all questions in this assignment.

The database is posted on Canvas

The E/R diagram for this database is below:



- 1) Find the number of requests per car-manufactured-year and gender. Are men more interested in newer cars than older cars?
- 2) Identify the oldest customer(s). List the name(s) and age(s). Try this with the ALL operator.

- 3) Among the customers who requested a ride to the Park, how many of them have names starting with the character 'S'?
- 4) List all customers, except the youngest one. Try this with the ANY operator.
- 5) Find the drivers whose average distance driven is larger than the average distance of all rides taken by Clayton residents. Sort the results by the average distance.
- 6) Find the number of requests per gender per destination. Select only destinations that have the character 't' in them. Sort results by destination then gender.
- 7) List the customers who had a ride to WashU in a silver car. Among these customers, who is the youngest? Use a subquery in the WHERE clause in your answer. Sort the results by age.
- 8) List the customers who are younger than the average age. Also, list those customers that all their rides are shorter than the average distance of all trips in any individual ride. Show all of these customers in one query. Sort the results by customer name in increasing order.
- 9) Find the customers who have more than the average number of requests in the current week. List their names and number of trips in the current week. Sort by the number of trips.
- 10) Among the customers in the previous query, which ones do not live in 'St. Louis'? (Hint: Use NOT IN)
- 11) Please add a new column to the requests table in order to identify the distance of rides and call it Trip. Please tag rides based on the following condition:
 - If the distance is larger than 20, tag it as "Long".
 - If the distance is between and including 15 and 20, tag it as "Upper Medium".
 - If the distance is between and including 10 and 14, tag it as "Medium".
 - If the distance is between and including 5 and 9, tag it as "Lower Medium".
 - If the distance is less than 5 tag it as "Short".
 - If no information is available for distance, please tag it as "No Information"
- 12) Please remove the column that you added recently (Trip).
- 13) List once each car model for cars that were requested on Tuesdays. Try this with the EXIST operator.